

In the previous section, we introduced the fundamentals of Number Theory. Now, we will move on to abstract algebra, which forms the core of modern cryptography. But you might wonder: why should we study abstract algebra?

Consider the set $\mathbb{Z}_n$. Naturally, we want to define certain arithmetical operations over $\mathbb{Z}_n$. For example, having $3, 6 \in \mathbb{Z}_9$ I would like to know what it means to add them. Naturally, one would say that $3 + 6 = 0$ in $\mathbb{Z}_9$ since $3 + 6$ over $\mathbb{Z}$ gives 9, which is congruent to 0 modulo 9. However, while such a structure is straightforward for such a set, we can define operations over much more difficult sets such as elliptic curve points, which are fundamental blocks for many Cryptography protocols.

In a way, abstract algebra gives *interfaces* to objects, just like in programming languages, where interfaces specify certain common patterns observed in different instances of this interface.

Let us explore this concept on the previously defined set $\mathbb{Z}_n$. We present the following lemma to show the properties of modular arithmetic.

Lemma 0.1. Suppose $a, b, c \in \mathbb{Z}_n$, then we have the following modular addition properties,

1. *Addition associative:* $a + (b + c) \equiv (a + b) + c \pmod{n}$. The order in which we compute pairwise sums does not matter.
2. *Addition identity:* $a + 0 \equiv a \pmod{n}$. We always have an element which behaves like “zero” in the addition operation (or like “one” when multiplying).
3. *Additive inverse:* $a + (-a) \equiv 1 \pmod{n}$. For any element a we always have an element $-a$ which “zeroes out” an element a .
4. *Addition commutative:* $a + b \equiv b + a \pmod{n}$. The order of elements in the addition operation does not matter.

Remark. All these properties also hold for multiplication operation, but with one drastic exception: the existence of the inverse element. So in other words, the multiplicative inverse property does not hold in $\mathbb{Z}_{16}$.

In abstract algebra, we explore algebraic structures defined by abstract operations. These structures possess specific properties that help us understand the behavior of their elements. By examining these properties, we can derive general principles and insights applicable to any mathematical systems.

0.1 Groups

Let us jump straight to the most fundamental algebraic structure – the group. Based on the introduction above, based on the set $\mathbb{G}$, we want some operation $\oplus$ to have certain properties, which are specified in the following definition.

Definition 0.2. A **group**, denoted by $(\mathbb{G}, \oplus)$, is a set with a binary operation $\oplus$, obeying the following rules:

1. **Closure:** Binary operations always output an element from $\mathbb{G}$, that is $\forall a, b \in \mathbb{G} : a \oplus b \in \mathbb{G}$.
2. **Associativity:** $\forall a, b, c \in \mathbb{G} : (a \oplus b) \oplus c = a \oplus (b \oplus c)$.
3. **Identity element:** There exists a so-called identity element $e \in \mathbb{G}$ such that $\forall a \in \mathbb{G} : e \oplus a = a \oplus e = a$.
4. **Inverse element:** $\forall a \in \mathbb{G} \exists b \in \mathbb{G} : a \oplus b = b \oplus a = e$. We commonly denote the inverse element by $(\ominus a)$ or a^{-1} .

Quite confusing at first glance, right? The best way to grasp this concept is to consider a couple of examples.

Example 0.1. A group of integers with the regular addition $(\mathbb{Z}, +)$ (also called the *additive group of integers*) is a group. Indeed, an identity element is $e_{\mathbb{Z}} = 0$, associativity obviously holds, and an inverse for each element $a \in \mathbb{Z}$ is $(\ominus a) := -a \in \mathbb{Z}$.

Remark. We use the term **additive group** when we mean that the binary operation is addition $+$, while **multiplicative group** means that we are multiplying two numbers via $\times^a$.

^aIn this section, regard $\cdot$ and $\times$ as the same operation of multiplication.

Example 0.2. The multiplicative group of positive real numbers $(\mathbb{R}_{>0}, \times)$ is a group for similar reasons. An identity element is $e_{\mathbb{R}_{>0}} = 1$, while the inverse for $a \in \mathbb{R}_{>0}$ is defined as $\frac{1}{a}$.

Example 0.3. The additive set of natural numbers $(\mathbb{N}, +)$ is not a group. Although operation of addition is closed, there is no identity element nor inverse element for, say, 2 or 10.

Exercise. Prove that for any group $\mathbb{G} = \{g, e\}$ with two elements and identity element e it holds that $g^2 = e$.

One might ask a reasonable question: suppose you pick $a, b \in \mathbb{G}$. Is $a \oplus b$ the same as $b \oplus a$? Unfortunately, for some groups, this is not true.

For this reason, it makes sense to give a special name to a group in which the operation is commutative (meaning, we can swap the elements).

Definition 0.3. A group $(\mathbb{G}, \oplus)$ is called **abelian** if $\forall a, b \in \mathbb{G} : a \oplus b = b \oplus a$.

Example 0.4. The additive group of integers $(\mathbb{Z}, +)$ is an abelian group. Indeed, $a + b = b + a$ for any $a, b \in \mathbb{Z}$.

Example 0.5. The set of 2×2 matrices with real entries and determinant 1 (denoted by $\text{SL}(2, \mathbb{R})$) is a group with respect to matrix multiplication. However, this group is not abelian! Take

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

Then, it is easy to verify that

$$AB = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad BA = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix},$$

so clearly $AB \neq BA$ – the elements of $\text{SL}(2, \mathbb{R})$ do not commute.

Remark. Further, we will write ab instead of $a \times b$ and a^{-1} instead of $\ominus a$ for the sake of simplicity (and because it is more common in the literature). As mentioned before, it is usually called the *multiplicative notation*.

Finally, for cryptography it is important to know the number of elements in a group. This number is called the *order* of the group.

Definition 0.4. The **order** of a finite group $\mathbb{G}$ is the number of elements in the group. We denote the order of a group as $|\mathbb{G}|$.

Example 0.6. Integers modulo 13, denoted by $\mathbb{Z}_{13}$, is a group with respect to addition modulo 13 (e.g., $5 + 12 = 4$ in $\mathbb{Z}_{13}$). The order of this group is 13.

Despite the aforementioned definitions, many things are not generally obvious. For example, one might ask whether the identity element is unique. Or, whether the inverse element is unique for each group element. For that reason, we formulate the following lemma.

Lemma 0.5. Suppose $\mathbb{G}$ is a group. Then, the following statements hold:

1. The identity element is unique.
2. The inverse element is unique for each element: $\forall a \in \mathbb{G} \exists! a^{-1} \in \mathbb{G} : aa^{-1} = a^{-1}a = e$.
3. For all $a, b \in \mathbb{G}$ there is a unique $x \in \mathbb{G}$ such that $ax = b$.
4. If $ab = ac$ then $b = c$. Similarly, if $xy = zy$ then $x = z$.

Since this guide is not a textbook on abstract algebra, we will not prove all the statements. However, we will prove the first and second one to show the nature

of the proofs in abstract algebra.

Proof (first statement). ► Suppose we have two identity elements $e_1, e_2 \in \mathbb{G}$. Consider their product $e_1 e_2$. By the identity element definition, we know that $e_1 e_2 = e_1$ and $e_1 e_2 = e_2$. It follows immediately that $e_1 = e_2$. ◀

Proof (second statement). ► Take some element $g \in \mathbb{G}$ and suppose $a, b \in \mathbb{G}$ are both inverses of g . By the inverse element definition, we have

$$ag = ga = e, \quad bg = gb = e.$$

Now, notice the following:

$$a = ae = a(gb) = (ag)b = eb = b$$

Thus, we have proven that $a = b$. ◀

Exercise. Prove the third and fourth statements.

0.2 Subgroups

When we are finally comfortable with the concept of a group, we can move on to the concept of a *subgroup*.

Suppose we have a group $(\mathbb{G}, \oplus)$. Suppose one takes the subset $\mathbb{H} \subset \mathbb{G}$. Of course, since all elements in $\mathbb{H}$ are still elements in $\mathbb{G}$, we can conduct operations between them via $\oplus$. The natural question to ask is whether $\mathbb{H}$ is a group itself. We say that $\mathbb{H}$ is a **subgroup** of $\mathbb{G}$ if this is indeed the case.

Definition 0.6. A subset $\mathbb{H} \subset \mathbb{G}$ is called a **subgroup** of $\mathbb{G}$ if $\mathbb{H}$ is a group with respect to the same operation $\oplus$. We denote this as $\mathbb{H} \leq \mathbb{G}$.

Example 0.7. Of course, not every subset of $\mathbb{G}$ is a subgroup. Take $(\mathbb{Z}, +)$. If we cut, say, 3 out of $\mathbb{Z}$ (so we get $\mathbb{H} = \mathbb{Z} \setminus \{3\}$), then $\mathbb{H}$ is not a subgroup of $\mathbb{Z}$ since an element -3 does not have an inverse in $\mathbb{H}$. Moreover, it is not closed: take $1, 2 \in \mathbb{H}$. In this case, $1 + 2 = 3 \notin \mathbb{H}$.

Example 0.8. Now, let us define some valid subgroup of $\mathbb{Z}$. Take $\mathbb{H} = 3\mathbb{Z}$. This is a subgroup of $\mathbb{Z}$, since it is closed under addition, has an identity element 0, and has an inverse for each element $3k$ (namely, $-3k$). That being said, $\mathbb{H} = 3\mathbb{Z} \leq \mathbb{Z}$.

0.3 Cyclic Groups

Cyclic groups are the most interesting groups in cryptography because of their simple yet powerful structure. In a cyclic group, every element can be created by repeatedly applying the group operation to one element, called the *generator*. This makes cyclic groups useful for many cryptographic algorithms, allowing for secure and efficient operations.

However, before diving deeper, we need to know how to multiply (add) elements multiple times. Traditionally, cyclic groups use the multiplicative notation, so we stick to it. Suppose we have a group $(\mathbb{G}, \times)$ and $g \in \mathbb{G}$. Then, for positive n we define g^n to be the result of multiplying g by itself n times. For $n = 0$ we assume $g^0 = 1$ with $1 \in \mathbb{G}$ being the identity element. Finally, for negative n we define $g^{-n} = (g^{-1})^n$. Now, let us consider the first definition.

Definition 0.7. Let $\mathbb{G}$ be a group and $g \in \mathbb{G}$, then **cyclic subgroup** generated by g is $\langle g \rangle = \{g^n : n \in \mathbb{Z}\} = \{\dots, g^{-3}, g^{-2}, g^{-1}, e, g, g^2, g^3, \dots\}$.

Example 0.9. Let $\mathbb{Z}_{12}$ be the group of integers modulo 12. Consider element $2 \in \mathbb{Z}_{12}$ then the group generated by 2 is

$$\langle 2 \rangle = \{2, 4, 6, 8, 10, 0\}$$

Note that the set $\langle g \rangle = \{g^n : n \in \mathbb{Z}\}$ is indeed a subgroup of $\mathbb{G}$. Indeed, it is closed under multiplication, has an identity element $e \in \langle g \rangle$ (corresponding to g^0), and has an inverse for each element (since for each $g^n \in \langle g \rangle$ the inverse is defined as $g^{-n} \in \langle g \rangle$).

For finite groups $\mathbb{G}$ we naturally expect $\langle g \rangle$ to have a finite size. Such size is what we call the *order* of the element g .

Definition 0.8. Let $\mathbb{G}$ be a group and $g \in \mathbb{G}$. The minimal natural number n for which $g^n = 1$ is called the **order** of element g and denote it by $\text{ord}(g)$. If such integer does not exist, we assume that $\text{ord}(g) = \infty$.

Example 0.10. Suppose we have a group $(\mathbb{Z}_9, +)$. Let us find the order of element $a = 3$. It is easy to notice that $3 \cdot 3 \equiv 0 \pmod{9}$, thus $\text{ord}(3) = 3$.

Example 0.11. Let $(\mathbb{Z}, +)$ be a group. Any non-zero integer $a \in \mathbb{Z}$ has an infinite order (while 0 has an order of 1).

Finally, we can define what it means for a group to be cyclic.

Definition 0.9. We say that a group $\mathbb{G}$ is **cyclic** if there exists an element $g \in \mathbb{G}$ such that $\mathbb{G}$ is generated by g , that is, $\mathbb{G} = \langle g \rangle$. In this case, g is a generator of $\mathbb{G}$.

Example 0.12. The group of integers $(\mathbb{Z}, +)$ is an infinite cyclic group. Indeed, it is generated by 1.

Now it is time to discuss some interesting properties of cyclic groups and their subgroups. We start with the following theorem.

Theorem 0.10. Every cyclic group is abelian.

Reasoning. Suppose the generator of the group is g . Then, for any $a, b \in \mathbb{G}$ we have

$$ab = g^m g^n = g^{m+n} = g^{n+m} = g^n g^m = ba.$$

Cryptographic protocols typically rely on the properties of cyclic groups, so it turns out that they usually work with abelian groups by default. Common cryptographic groups, like those based on elliptic curves or modular arithmetic, are cyclic and, therefore, abelian.

Lemma 0.11. Let $\mathbb{G} = \langle g \rangle$ be a cyclic group of order n . Then $g^m = e$ holds if and only if $m \mid n$.

Theorem 0.12. Let $\mathbb{G} = \langle g \rangle$ be a cyclic group of order n . If $h = g^m$, then the order of h is $n/\gcd(m, n)$.

Consider this theorem as a generalization of [Definition 0.11](#). By using the above theorem, we can derive a criterion which the generators of $\mathbb{Z}_n$ must adhere to.

Corollary 0.13. The generators of group $\mathbb{Z}_n$ are the integers from $\mathbb{Z}_n^\times$.

It is also easy to see that the order of any group generator is the same as the group it generates.

Example 0.13. Let us examine how many elements can generate $(\mathbb{Z}_{12}, +)$. By theorem above each element of $\mathbb{Z}_{12}^\times = \{1, 5, 7, 11\}$ generates $\mathbb{Z}_{12}$. For example, let us consider $\langle 5 \rangle$:

• $5^1 = 0 + 5 = 5$	• $5^5 = 8 + 5 = 1$	• $5^9 = 4 + 5 = 9$
• $5^2 = 5 + 5 = 10$	• $5^6 = 6 + 5 = 6$	• $5^{10} = 2 + 5 = 7$
• $5^3 = 10 + 5 = 3$	• $5^7 = 11 + 5 = 11$	• $5^{11} = 7 + 5 = 7$
• $5^4 = 3 + 5 = 8$	• $5^8 = 4 + 5 = 4$	• $5^{12} = 7 + 5 = 0$

As we see, all elements of $\mathbb{G}$ are generated by the element 5. However, does it hold for any other integer from $\mathbb{Z}_n \setminus \mathbb{Z}_n^\times$? Consider $\langle 3 \rangle$ as an example:

• $3^1 = 3$	• $3^2 = 6$	• $3^3 = 9$	• $3^4 = 0$
-------------	-------------	-------------	-------------

While this is a valid subgroup, it does not generate the whole group $(\mathbb{Z}_{12}, +)$.

Based on the previous example, you may have a reasonable question about the types of subgroups of cyclic groups. For example, can we find some non-cyclic subgroup of $\mathbb{Z}_{12}$? Turns out that this is not possible due to the subsequent

theorem.

Theorem 0.14. Every subgroup of cyclic group is cyclic.

0.4 Cosets and Lagrange's Theorem*

Lagrange's Theorem is one of the most important results in finite group theory, considering all the consequences. Central to understanding Lagrange's Theorem is the notion of a *coset*. While this concept is essential for Cryptography, feel free to skip this section if feel uncomfortable with the material.

Definition 0.15. Let $\mathbb{G}$ be a group and $\mathbb{H}$ a subgroup of $\mathbb{G}$. **Left coset** with representative $g \in \mathbb{G}$ is defined as $g\mathbb{H} = \{gh : h \in \mathbb{H}\}$. Similarly, the **right coset** with representative $g \in \mathbb{G}$ is defined as $\mathbb{H}g = \{hg : h \in \mathbb{H}\}$.

Example 0.14. Suppose $\mathbb{H}$ be subgroup of $\mathbb{Z}_8$ with elements $\{0, 4\}$. Let us find all the cosets

$$0 + \mathbb{H} = 4 + \mathbb{H} = \{0, 4\}$$

$$1 + \mathbb{H} = 5 + \mathbb{H} = \{1, 5\}$$

$$2 + \mathbb{H} = 6 + \mathbb{H} = \{0, 6\}$$

$$3 + \mathbb{H} = 7 + \mathbb{H} = \{3, 7\}$$

Note that in a commutative group, left and right cosets are always identical, so for instance $4 + \mathbb{H} = \mathbb{H} + 4$.

Lemma 0.16. For any group element $g \in \mathbb{G}$ it holds that $|g\mathbb{H}| = |\mathbb{H}|$.

Definition 0.17. Let $\mathbb{G}$ be a group and $\mathbb{H}$ be a subgroup of $\mathbb{G}$. The **index** of subgroup $\mathbb{H}$ in group $\mathbb{G}$ is the number of left cosets of $\mathbb{H}$ in $\mathbb{G}$. We will denote this index by $[\mathbb{G} : \mathbb{H}]$.

Example 0.15. Let $\mathbb{G} = \mathbb{Z}_8$ and $\mathbb{H} = 4\mathbb{Z}_8 = \{0, 4\}$. Then, $[\mathbb{G} : \mathbb{H}] = 4$. Moreover, for every $m \in \mathbb{N}$ it holds that $[\mathbb{Z} : m\mathbb{Z}] = m$.

Finally, we are ready for Lagrange's Theorem. This is a fundamental result in the group theory, crucial for understanding the structure of finite groups.

Theorem 0.18 (Lagrange). If $\mathbb{H}$ is a subgroup of an any finite group $\mathbb{G}$, then $|\mathbb{G}| = [\mathbb{G} : \mathbb{H}] |\mathbb{H}|$.

Proof. ► The group $\mathbb{G}$ is partitioned into $[\mathbb{G} : \mathbb{H}]$ distinct left cosets. Each

left coset has $|\mathbb{H}|$ elements, therefore $|\mathbb{G}| = [\mathbb{G} : \mathbb{H}] |\mathbb{H}|$. ◀.

Corollary 0.19. Suppose $\mathbb{H}$ is subgroup of a finite group $\mathbb{G}$, then $|\mathbb{H}| \mid |\mathbb{G}|$.

Proof. ▶ By Lagrange's theorem, the order of the group $\mathbb{G}$ decomposes as the product of the index $[\mathbb{G} : \mathbb{H}]$ and the order of the subgroup $\mathbb{H}$. Therefore, by the definition of divisibility (see ??), it follows directly that $|\mathbb{H}|$ divides $|\mathbb{G}|$ ◀.

Corollary 0.20. Let $\mathbb{G}$ be a finite group and $g \in \mathbb{G}$, then $\text{ord}(g) \mid |\mathbb{G}|$. In other words, the order of g must divide the numbers of elements in $\mathbb{G}$.

Additionally, consider the following fact which seems to be obvious but required aforementioned theorems to be easily proven.

Corollary 0.21. Let $\mathbb{G}$ be a group of order n and $g \in \mathbb{G}$. Then, $g^n = e$.

Example 0.16. Let us take the multiplicative group $\mathbb{Z}_{17}^\times$. As we already know, the order of this group is $\varphi(17) = 16$. So, for any element $x \in \mathbb{Z}_{17}^\times$ it holds that $x^{16} = 1$. Note that this fact is also a direct consequence of the Euler's theorem: $x^{\varphi(n)} \equiv 1 \pmod{n}$.

Based on the example above, one might notice that **Definition 0.21** serves as a generalization of Euler's Theorem for arbitrary finite group.

0.5 Isomorphisms and Endomorphisms

Finally, we will define the concept of isomorphisms and endomorphisms. These are important concepts in the world of cryptography, since they allow us to compare different groups. Namely, suppose we have two groups $(\mathbb{G}, \oplus)$ and $(\mathbb{H}, \odot)$. Is there any way to state that these two groups are the same? The answer is yes, and this is done via isomorphisms.

Definition 0.22. A function $\varphi : \mathbb{G} \rightarrow \mathbb{H}$ is called an **homomorphism** if it is a function that preserves the group operation, that is,

$$\forall a, b \in \mathbb{G} : \varphi(a \oplus b) = \varphi(a) \odot \varphi(b).$$

Definition 0.23. An **isomorphism** is a bijective homomorphism.

Definition 0.24. If there exists an isomorphism between two groups $\mathbb{G}$ and $\mathbb{H}$, we say that these groups are isomorphic and write $\mathbb{G} \cong \mathbb{H}$.

Example 0.17. Consider the group of integers $(\mathbb{Z}, +)$ and the group of integers modulo 12 $(\mathbb{Z}_{12}, +)$. The function $\varphi : \mathbb{Z} \rightarrow \mathbb{Z}_{12}$ defined as $\varphi(x) = x \bmod 12$ is a homomorphism. Indeed:

$$\varphi(a + b) = (a + b) \bmod 12 = (a \bmod 12) + (b \bmod 12) = \varphi(a) + \varphi(b).$$

However, this function is not an isomorphism, since it is not bijective. For example, $\varphi(0) = \varphi(12) = 0$.

Example 0.18. Additive group of reals $(\mathbb{R}, +)$ and the multiplicative group of positive reals $(\mathbb{R}_{>0}, \times)$ are isomorphic. The function $\varphi : \mathbb{R} \rightarrow \mathbb{R}_{>0}$ defined as $\varphi(x) = e^x$ is an isomorphism. Indeed:

$$\varphi(a + b) = e^{a+b} = e^a \cdot e^b = \varphi(a) \cdot \varphi(b).$$

Thus, φ is a homomorphism. It is also injective since $e^x = e^y \implies x = y$. Finally, it is obviously onto. This means $(\mathbb{R}, +) \cong (\mathbb{R}_{>0}, \times)$.

Example 0.19. All groups of order 2 are isomorphic to $\mathbb{Z}_2$. Indeed, let $\mathbb{G} = \{g, e\}$ – any group of order 2, and define $\varphi : \mathbb{Z}_2 \rightarrow \mathbb{G}$ as $\varphi(0) = e$ and $\varphi(1) = g$. This is an isomorphism.

A generalization of the above example is the following interesting theorem.

Theorem 0.25. Suppose $\mathbb{G} = \langle g \rangle$ is a finite cyclic group and $|G| = n \in \mathbb{N}$. Then, $\mathbb{G} \cong \mathbb{Z}_n$.

Idea of the proof. Define a function $\varphi : \mathbb{Z}_n \rightarrow \mathbb{G}$ as $m \mapsto g^m$. One can prove that this is an isomorphism.

Here, it is quite evident that isomorphism tells us that the groups have the same structure. Moreover, it is correct to say that if $\mathbb{G} \equiv \mathbb{H}$, then $\mathbb{G}$ and $\mathbb{H}$ are *equivalent* since $\cong$ is an equivalence relation.

Exercise (*). Prove that $\cong$ is an equivalence relation.

Finally, we will define the concept of an endomorphism and automorphism to finish the section.

Definition 0.26. An **endomorphism** is a function φ which maps set X to itself ($\varphi : X \rightarrow X$).

Definition 0.27. An **automorphism** is an isomorphic endomorphism.

Example 0.20. Given a group $\mathbb{G}$, fixate $a \in \mathbb{G}$. The map $\varphi : x \mapsto axa^{-1}$ is an automorphism.

Last two definitions are especially frequently used in Elliptic Curves theory.

0.6 Fields

Although typically one introduces rings before fields, we believe that for the basic understanding, it is better to start with fields.

Notice that when dealing with groups, we had a single operation $\oplus$, which, depending on the context, is either interpreted as addition or multiplication. However, fields allow to extend this concept a little bit further by introducing a new operation, say, $\odot$, which, combined with $\oplus$, allows us to perform the basic arithmetic.

This is very similar to the real or rational numbers, for example. We can add, subtract, multiply, and divide them. This is exactly what fields are about, but in a more abstract way. That being said, let us see the definition.

Definition 0.28. A **field** is a set $\mathbb{F}$ with two operations $\oplus$ and $\odot$ such that:

1. $(\mathbb{F}, \oplus)$ is an abelian group with identity $e_{\oplus}$.
2. $(\mathbb{F} \setminus \{e_{\oplus}\}, \odot)$ is an abelian group.
3. The **distributive law** holds:

$$\forall a, b, c \in \mathbb{F} : a \odot (b \oplus c) = (a \odot b) \oplus (a \odot c).$$

What this definition states is that we can perform the following operations:

1. Addition: $a \oplus b$, inherited from group structure $(\mathbb{F}, \oplus)$.
2. Subtraction: $a \oplus (\ominus b)$, inherited from group structure $(\mathbb{F}, \oplus)$.
3. Multiplication: $a \odot b$, inherited from group structure $(\mathbb{F} \setminus \{e_{\oplus}\}, \odot)$.
4. Division: $a \odot b^{-1}$, except for $b = 0$, inherited from group structure $(\mathbb{F} \setminus \{e_{\oplus}\}, \odot)$.

Example 0.21. The set of real numbers $(\mathbb{R}, +, \times)$ is obviously a field.

Example 0.22. The set of complex numbers $(\mathbb{C}, +, \times)$ is also a field. Indeed, let us see how we can perform operations. Suppose we are given $z = a_0 + a_1 i$ and $w = b_0 + b_1 i$ with $i^2 + 1 = 0$. In this case:

1. Addition: $z + w = (a_0 + b_0) + (a_1 + b_1)i$.
2. Subtraction: $z - w = (a_0 - b_0) + (a_1 - b_1)i$.
3. Multiplication: $z \cdot w = (a_0 b_0 - a_1 b_1) + (a_0 b_1 + a_1 b_0)i$.
4. Division: $z/w = \frac{a_0 b_0 + a_1 b_1}{b_0^2 + b_1^2} + \frac{a_1 b_0 - a_0 b_1}{b_0^2 + b_1^2} i$.

Interestingly though, it is very difficult to come up with some more complicated, non-trivial examples. For that reason, we will simply move to the most central field used in cryptography – finite fields.

0.7 Finite Fields

Notice that $(\mathbb{Z}, +, \times)$ does not form a field since division is not closed. Similarly, the tuple $(\mathbb{Z}_n, +, \times)$ also does not always form a field since the multiplicative inverses are defined only for coprime numbers. This motivates us to consider the set $\mathbb{Z}_p$ with prime p in which every integer has a well-defined inverse modulo p . Such fields are called *prime fields*.

Definition 0.29. The **Prime field** of order p is a set $\{0, 1, \dots, p-2, p-1\}$, in which operations are defined “modulo p ” (see details above). Typically, finite fields are denoted as $\mathbb{F}_p$ or $\text{GF}(p)$.

Finite fields is the core object in cryptography. Instead of real numbers or pure integers, we will almost always use finite fields.

Remark. In many cases, one might encounter both $\mathbb{F}_p$ and $\mathbb{Z}_p$ notations. The difference is the following: when one refers to $\mathbb{Z}_p$, it is typically assumed that the operations are performed in the ring^a of integers modulo p (meaning, we need only addition, subtraction, and multiplication in the protocol), while division is of little interest. When one refers to $\mathbb{F}_p$, it is typically assumed that we need full arithmetic (including division) for the protocol.

^aWe have not defined as of now what ring is, but, roughly speaking, this is a field without multiplicative inverses (such as $\mathbb{Z}_n$ for composite n)

Example 0.23. Consider $9, 14 \in \mathbb{F}_{17}$. Some examples of calculations:

1. $9 + 14 = 6$.
2. $9 - 14 = 12$.
3. $9 \times 14 = 7$.
4. $14^{-1} = 11$ since $14 \cdot 11 = 154 \equiv 1 \pmod{17}$.

0.8 Some Fun: Group Implementation in Rust

In programming, we can think of a group as an interface, having a single binary operation defined, that obeys the rules of closure, associativity, identity element, and inverse element.

For that reason, we might even code a group in Rust! We will also write a simple test to check whether the group is valid and whether the group is abelian.

Trait for Group. First, we define a trait for a group. We will define a group as a trait with the following methods:

```
/// Trait that represents a group.
pub trait Group: Sized {
    /// Checks whether the two elements are equal.
    fn eq(&self, other: &Self) -> bool;
```

```

    /// Returns the identity element of the group.
    fn identity() -> Self;
    /// Adds two elements of the group.
    fn add(&self, a: &Self) -> Self;
    /// Returns the negative of the element.
    fn negate(&self) -> Self;
    /// Subtracts two elements of the group.
    fn sub(&self, a: &Self) -> Self {
        self.add(&a.negate())
    }
}

```

Checking group validity. Now observe the following: we get closure for free, since the compiler will check whether the return type of the operation is the same as the type of the group. However, there is no guarantee that associativity holds, and our identity element is at all valid. For that reason, we need to somehow additionally check the validity of implementation.

We propose to do the following: we will randomly sample three elements from the group $a, b, c \leftarrow \mathbb{G}$ and check our three properties:

1. $a \oplus (b \oplus c) \stackrel{?}{=} (a \oplus b) \oplus c.$
2. $a \oplus e \stackrel{?}{=} e \oplus a \stackrel{?}{=} a.$
3. $a \oplus (\ominus a) \stackrel{?}{=} (\ominus a) \oplus a \stackrel{?}{=} e.$

Additionally, if we want to verify whether the group is abelian, we can check whether $a \oplus b \stackrel{?}{=} b \oplus a.$

For that reason, for the check, we require the group to be samplable (i.e. we can randomly sample elements from the group):

```

/// Trait for sampling a random element from a group.
pub trait Samplable {
    /// Returns a random element from the group.
    fn sample() -> Self;
}

```

And now, our test looks as follows:

```

/// Number of tests to check the group properties.
const TESTS_NUMBER: usize = 100;

/// Asserts that the given group G is valid.
/// A group is valid if the following properties hold:
/// 1. Associativity: (a + b) + c = a + (b + c)
/// 2. Identity: a + e = a = e + a
/// 3. Inverse: a + (-a) = e = (-a) + a
pub fn assert_group_valid<G>()
where
    G: Group + Samplable,

```

```

{
  for _ in 0..TESTS_NUMBER {
    // Take random three elements
    let a = G::sample();
    let b = G::sample();
    let c = G::sample();

    // Check whether associativity holds
    let ab_c = a.add(&b).add(&c);
    let a_bc = a.add(&b.add(&c));
    let associativity_holds = ab_c.eq(&a_bc);
    assert!(associativity_holds, "Associativity does
    ↪ not hold for the given group");

    // Check whether identity element is valid
    let e = G::identity();
    let ae = a.add(&e);
    let ea = e.add(&a);
    let identity_holds = ae.eq(&a) && ea.eq(&a);
    assert!(identity_holds, "Identity element does
    ↪ not hold for the given group");

    // Check whether inverse element is valid
    let a_neg = a.negate();
    let a_neg_add_a = a_neg.add(&a);
    let a_add_a_neg = a.add(&a_neg);
    let inverse_holds = a_neg_add_a.eq(&e) &&
    ↪ a_add_a_neg.eq(&e);
    assert!(inverse_holds, "Inverse element does not
    ↪ hold for the given group");
  }
}

/// Asserts that the given group G is abelian.
/// A group is an abelian group if the following
  ↪ property holds:
///  $a + b = b + a$  for all  $a, b$  in  $G$  (commutativity)
pub fn assert_group_abelian<G>()
where
  G: Group + Samplable,
{
  for _ in 0..TESTS_NUMBER {
    assert_group_valid::<G>();

    // Take two random elements
    let a = G::sample();
    let b = G::sample();

```

```

        // Check whether commutativity holds
        let ab = a.add(&b);
        let ba = b.add(&a);
        assert!(ab.eq(&ba), "Commutativity does not hold
        ↪ for the given group");
    }
}

```

Testing the group ($\mathbb{Z}, +$). And now, we can define a group for integers and check whether it is valid and abelian:

```

use crate::group::{Group, Samplable};
use rand::Rng;

/// Implementing group for Rotation3<f32>
impl Group for i64 {
    fn eq(&self, other: &Self) -> bool {
        self == other
    }

    fn identity() -> Self {
        0i64
    }

    fn add(&self, a: &Self) -> Self {
        self + a
    }

    fn negate(&self) -> Self {
        -self
    }
}

impl Samplable for i64 {
    fn sample() -> Self {
        let mut gen = rand::thread_rng();

        // To prevent overflow, we choose a smaller
        ↪ range for i64
        let min = i64::MIN / 3;
        let max = i64::MAX / 3;
        gen.gen_range(min..max)
    }
}

```

Just a small note: since we cannot generate infinite integers, we restrict the range of integers to prevent overflow. So, for the sake of simplicity, we divide the range of integers by 3, in which overflow never occurs.

And now, the moment of truth! Let us define some tests and run them:

```
#[cfg(test)]
mod tests {
    use super::*;
    use group::*;

    #[test]
    fn test_integers_are_group() {
        assert_group_valid::<i64>()
    }

    #[test]
    fn test_integers_are_abelian() {
        assert_group_abelian::<i64>();
    }
}
```

Both tests pass! Now let us consider something a bit trickier.

Testing the group $SO(3)$. We can define a group for 3×3 rotation matrices. Of course, composition of two rotation is not commutative, so we expect the abelian test to fail. However, the group is still valid! For example, there is an identity rotation matrix E , and for each rotation matrix $A \in SO(3)$, there exists a rotation matrix $A^{-1} \in SO(3)$ such that $AA^{-1} = A^{-1}A = E$. Finally, the associativity holds as well.

We will use the `nalgebra` library for this purpose, which contains the implementation of rotation matrices. So our implementation can look as follows:

```
/// A threshold below which two floating point numbers
    ↪ are considered equal.
const EPSILON: f32 = 1e-6;

/// Implementing group for Rotation3<f32>
impl Group for Rotation3<f32> {
    fn eq(&self, other: &Self) -> bool {
        // Checking whether the norm of a difference is
        ↪ small
        let difference = self.matrix() - other.matrix();
        difference.norm_squared() < EPSILON
    }

    fn identity() -> Self {
        Rotation3::identity()
    }

    fn add(&self, a: &Self) -> Self {
        self * a
    }
}
```

```

    fn negate(&self) -> Self {
        self.inverse()
    }
}

impl Samplable for Rotation3<f32> {
    fn sample() -> Self {
        let mut gen = rand::thread_rng();

        // Pick three random angles
        let roll = gen.gen_range(0.0..1.0);
        let pitch = gen.gen_range(0.0..1.0);
        let yaw = gen.gen_range(0.0..1.0);

        Rotation3::from_euler_angles(roll, pitch, yaw)
    }
}

```

Here, there are two tricky moments:

1. We cannot compare floating point numbers directly, since they might differ by a small amount. For that reason, we define a small threshold ε . We say that two matrices are equal iff the norm¹ of their difference is less than ε .
2. To generate a random rotation matrix, we generate three random angles and create a rotation matrix from these angles.

Acknowledgements

This section was greatly inspired by [Michael Penn's "Abstract Algebra" Youtube series](#) and [MIT Modern Algebra Class \(18.703\)](#). For those interested to dive deeper into the topic, we highly recommend these resources.

0.9 Exercises

Exercise 1. Define $X := \{(x, y) \in \mathbb{Q}^2 : xy = 1\}$. Oleksandr claims that:

1. $X \cap \mathbb{N}^2 = \{(1, 1)\}$.
2. $|X \cap \mathbb{Z}^2| = 2|X \cap \mathbb{N}^2|$.
3. $(X, \odot)$ is a group for $(x_1, y_1) \odot (x_2, y_2) = (x_1x_2, y_1y_2)$.

Which statements are **true**?

- a) Only 1.
- b) Only 1 and 2.
- c) Only 1 and 3.

¹one can think of norm as being the measure of “distance” between two objects. Similarly, we can define norm not only on matrices, but on vectors as well.

- d) Only 2 and 3.
- e) All statements are correct.

Exercise 2. Does $(\mathbb{Z}, \oplus)$ with operation $a \oplus b = a + b - 1$ define a group?

- a) Yes, and this group is abelian.
- b) Yes, but this group is not abelian.
- c) No, since the associativity property does not hold.
- d) No, since there is no identity element in this group.
- e) No, since there is no inverse for some elements in this group.

Exercise 3. Consider the Cartesian plane $\mathbb{R}^2$, where two coordinates are real numbers. For two points A, B define the operation $\oplus$ as follows: $A \oplus B$ is the midpoint on segment AB . Does $(\mathbb{R}^2, \oplus)$ define a group?

- a) Yes, and this group is abelian.
- b) Yes, but this group is not abelian.
- c) No, since the associativity property does not hold and there is no identity element in this group.
- d) No, since the associativity property does not hold, but we might define an identity element nonetheless.

Exercise 4. Let $\mathbb{G}$ be the set of 2×2 matrices with integer entries and a non-zero determinant. Does $(\mathbb{G}, \times)$ form a group?

- a) Yes, and this group is abelian.
- b) Yes, but this group is not abelian.
- c) No, since the associativity property does not hold.
- d) No, since there is no identity element in this group.
- e) No, since there is no inverse for some elements in this group.

Exercise 5*. Let $(\mathbb{G}, \times)$ be an abelian group and Ω be any set. Let $\mathcal{F}$ be the set of functions $f : \Omega \rightarrow \mathbb{G}$ with domain Ω and range $\mathbb{G}$. Define the operation $\star$ on $\mathcal{F}$ by $(f \star g)(\omega) = f(\omega) \times g(\omega)$. Is $(\mathcal{F}, \star)$ a group?

- a) Yes, and this group is abelian.
- b) Yes, but this group is not abelian.
- c) No, since the associativity property does not hold.
- d) No, since there is no identity element in this group.
- e) No, since there is no inverse for some elements in this group.

Exercise 6. Which of the following is a **valid** automorphism $f : X \rightarrow X$?

- a) $X = [0, 1], f : x \mapsto x^2$.
- b) $X = [0, 1], f : x \mapsto x + 1$.
- c) $X = \mathbb{R}_{>0}, f : x \mapsto (x - 1)^3$.
- d) $X = \mathbb{Q}_{>0}, f : x \mapsto \sqrt{x}$.

Exercise 7*. Denote by $\text{GL}(2, \mathbb{R})$ a set of 2×2 invertible matrices with real entries. Define two functions $\varphi : \text{GL}(2, \mathbb{R}) \rightarrow \mathbb{R}$:

$$\varphi_1 \left(\begin{bmatrix} a & b \\ c & d \end{bmatrix} \right) = ad - bc, \quad \varphi_2 \left(\begin{bmatrix} a & b \\ c & d \end{bmatrix} \right) = a + d$$

Den claims the following:

1. φ_1 is a group homomorphism between multiplicative groups $(\text{GL}(2, \mathbb{R}), \times)$ and $(\mathbb{R}, \times)$.
2. φ_2 is a group homomorphism between additive groups $(\text{GL}(2, \mathbb{R}), +)$ and $(\mathbb{R}, +)$.

Which of the following is **true**?

- a) Only statement 1 is correct.
- b) Only statement 2 is correct.
- c) Both statements 1 and 2 are correct.
- d) None of the statements is correct.

Exercise 8*. Introduce the set $\mathbb{M}$ as follows:

$$\mathbb{M} = \left\{ \begin{bmatrix} a & -b \\ b & a \end{bmatrix} : a, b \in \mathbb{R}, a^2 + b^2 \neq 0 \right\}$$

Suppose the function $\psi : \mathbb{C} \setminus \{0\} \rightarrow (\mathbb{M}, \times)$ is given by:

$$\psi(a + bi) = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$$

Is the function ψ an isomorphism?

- a) Yes, ψ is an isomorphism.
- b) No, ψ is not an isomorphism, but it is a homomorphism.
- c) ψ is neither an isomorphism nor a homomorphism.

Exercise 9*. Prove that a group of order 9 is abelian.